Transistor Laser: A Review

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I. Background and Motivation

Although the CMOS transistor has been the industry standard for decades, it operates purely as a standard electronic voltage-controlled switch. As scaling reaches its physical limits, the "RC delay" and parasitic heat generated by electron transport in metallic interconnects have become significant bottlenecks [1][2].

To address these limitations, the field has moved to Heterojunction Bipolar Transistors (HBTs), which make use of minority carrier injection with different semiconductor materials to attain high RF switching rates. The Transistor Laser (TL) is a sophisticated version of an HBT. It is a three-port circuit that combines the speed of logic of an HBT and the optical output of a semiconductor laser since both are made of the same material structure. In addition, through the addition of Multi-Quantum Wells (MQWs) to the base region, the TL transforms the loss by recombination current into a high-speed optical signal [1].

Integrating lasers with electronic drivers requires growing HBT layers over laser materials, followed by complex etching to isolate components, which increases fabrication costs and limits device density [2].

Although the TL provides a unified material structure, current designs have technical challenges. Shallow ridge structures have a high free-carrier absorption in the p-type base, and the Buried Heterostructures (BH) are costly to fabricate. Moreover, the majority of reported TLs demonstrate a catastrophic reduction in current gain (β) when lasing occurs, which is a significant impediment to their operation in the HBT-based amplification [1].

The motivation behind this report is to examine a novel InGaAlAs/InP deep ridge TL that will balance optical efficiency and electrical stability. This structure will preserve high current gain in the operation of the laser by using a large conduction band offset and a bottom layer of n-doped InAlAs cladding layer. By making a successful realization, dual functionality, monolithic integration of high-quality electrical drivers with high-speed lasers would be possible, which would offer a viable prospect to the next generation of integrated optoelectronic circuits.

II. Methodology

Crosslight PICS 3D software, which solves electrical, optical, and laser rate equations, is used. The drift-diffusion model with a thermionic-emission model has been used to accurately simulate carrier transport across heterojunctions, and subsequently, the lateral optical modes were determined by the effective-index method. The MQW gain was calculated using a 4 X 4 k-p band algorithm that includes the effects of valence-mixing and a Lorentz broadening functionality to capture realistic scattering times. The key process in the research methodology was the progressive optimization of the epitaxial architecture of the device to eliminate the gain-drop effect of conventional TLs [1].

A. Fabrication methods and device characterization

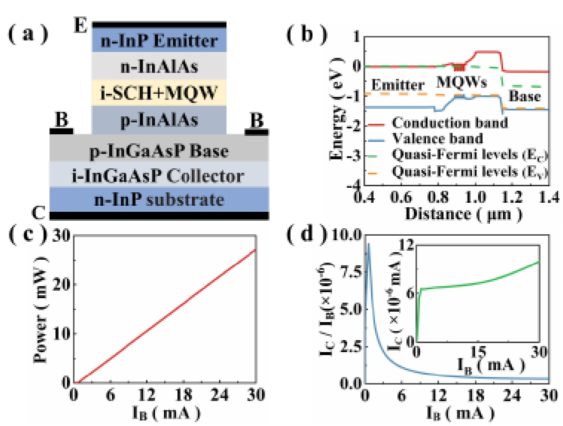

Figure 1 (a)Schematic structure of the TL having a p-InAlAs cladding layer, (b) band diagram at the center of the emitter ridge, (c) PI and (d) CE mode current gain of the TL. I_C as a function of I_B is shown in the inset of (d).

Figure 1 (a) shows a baseline model consisting of a p-doped InAlAs lower cladding layer, like a typical diode laser, which could easily achieve an optical power of 25 mW yet with a near-zero CE current gain of $9.4*10^{-6}$ as illustrated in Figure 1(b). This was the failure of this design as an energy barrier of the conduction band trapped the electrons in the MQW region.

To fix this, an n-doped InAlAs layer is placed just under the MQW as shown in Figure 2(a). This modification separated the laser and transistor modes, with the base current being fed through the n-InAlAs instead of being powered by emitter-injected electrons, which enabled a high current gain even when lasing was occurring, Figure 2(c) and (d). Moreover, the higher the level of doping of the n-InAlAs layer, the higher the CE current gain, and the lower the slope efficiency of the PI (optical power versus base current) curve.

Moreover, the gain was maximized by minimizing the electron loss to the base contact by reducing the p-base layer thickness to 75 nm. Lastly, to reduce the harsh effect of nonradiative recombination defects on the etched sidewalls, a selective oxidizing reaction was simulated. By forming a current confinement aperture within the InAlAs layers, the design funnels carriers through the center of the device, ensuring they recombine radiatively before diffusing to the defective surfaces, thereby drastically lowering the lasing threshold while preserving high electrical performance.

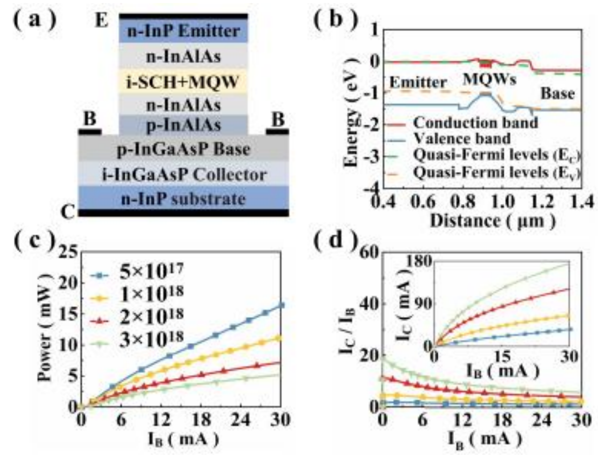

Figure 2 (a) Schematic structure of the TL having an n InAlAs + p-InAlAs lower cladding layer, (b) band diagram of the center of the emitter ridge, (c) PI and (d) CE mode current gain of the TL. The effects of the doping level in cm^{-3} of the n InAlAs cladding layer are shown in (c) and (d). For (b), the doping level of the InAlAs layer is 1 x 10^{18} cm^{-3}. I_C as a function of I_B is shown in the inset of (d).

III. RESULTS AND NOVELTY CLAIM

The most significant discovery of the study is that a historical constraint on TLs called the gain-drop problem was broken with the introduction of this design. In standard TL designs, the CE current gain (β) decreases sharply when the device enters its lasing threshold since the electrons injected by the emitter are now being used to create photons via stimulated recombination.

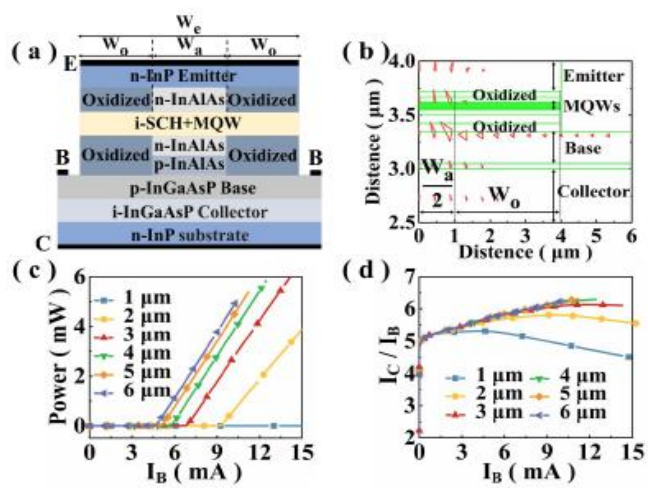

Figure 3 (a) Schematic structure of TL having n InAlAs +p-InAlAs cladding layer with current aperture formed by selective oxidation of InAlAs layers, (b) distribution of current in the cross section of the device. The PI and CE mode current gain (d) at different W_o are shown in (c) and (d), respectively. The doping level for the InAlAs layer in the device is 1 x 10^{18} cm^{-3}.

Such a strategic change in the carrier dynamics of the device essentially provides the high base current directly via the inserted n-InAlAs layer and essentially decouples the electrical output of the transistor to the optical output of the laser, which enables the device to stabilize a current gain of more than 4.3 with a simultaneous optical power output of 11.1 mW and a base current of 30 mA. It is worth noting that in this structure, the current gain does not reduce past the lasing threshold as is done in the past designs.

The current can be directed to the center of the device by using a selective oxidation aperture in the InAlAs layers to ensure that defective sidewalls are not exposed to the carriers and can reduce the threshold current to as low as 4.34 mA with a 6 µm oxidized aperture, Figure 3(c).

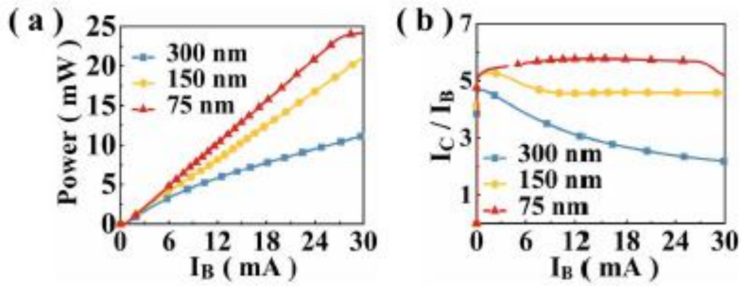

Figure 4 Optical power and current gain as a function of base current for different collector layer thicknesses.

Conversely, due to the fact that the n-doped InAlAs layer beneath the MQW supplies the collector current, the same device has a high current gain as well as emits light. As Figure 4 shows, when the p-base current is reduced to 10 mA, with the thickness of the base layer reduced to 75 nm and 300 nm, the power and current gain of the light increase to 8.3 mW and 5.8, respectively.

IV. Conclusion

In the case of this device, due to the n-InAlAs layer, a normal GRIN SCH (Graded Index Separate Confinement Heterostructure) structure, which assists in improving the device modulation and PI performance, can be embraced. In this way, it is possible to expect the fabrication of high-speed modulated lasers that are combined with high-quality electrical drivers [1].

V. References

[1] C. Zhang, X. Xiong, L. Qiao, M. Zhang, L. Zhao, and S. Liang, “Deep Ridge InGaAlAs/InP MQW Transistor Lasers,” *IEEE Electron Device Lett.*, vol. 45, no. 7, pp. 1225–1228, 2024, doi: 10.1109/LED.2024.3401780.

[2] Han Wui Then, M. Feng, and N. Holonyak, “The Transistor Laser: Theory and Experiment,” *Proceedings of the IEEE*, vol. 101, no. 10, 2013, doi: 10.1109/JPROC.2013.2274935.